

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

QIAN QI

ABSTRACT. We determine the finite-state prediction complexity of separated $C^{1+\gamma}$ expanding repellers carrying Hölder Gibbs measures. The optimal causal risk, allowing all randomized time-dependent observers, is comparable to a finite cylinder profile and to the quantization error of the weighted future orbit. Adaptive cylinder codes have suffix-closed full trees, with exactly $(bN - 1)/(b - 1)$ states for N leaves, yielding a stationary realization without a growing age factor. The squared-error exponent is computed by the larger root of two pressure equations, separating spatial distortion from survival through unobserved expansion. The result applies to nonconstant derivative cocycles and nonuniform Markov branch laws. A conditional minorization argument extends the risk laws to acquisition kernels that retain hidden state across observations; in particular it combines noisy acquisition with dependent expanding dynamics. Complete proofs of the general posterior-orbit and noisy expanding comparison theorems are included.

CONTENTS

1. Introduction	1
1.1. The principal results	2
1.2. Resource convention	2
1.3. Relation to the literature	3
1.4. Organization and dependencies	3
2. An intrinsic law on Gibbs repellers	4
2.1. A finite cylinder algorithm	5
2.2. The pressure law	5
3. Cylinder geometry and causal realization	6
4. Pressure, distortion and survival	9
4.1. Markov branch weights and unequal multipliers	11
5. Acquisitions with persistent hidden state	11
6. Posterior orbits and finite-state realization	14
7. Noisy expanding experiments and a uniform critical window	17
7.1. Geometric renewals and the critical window	19
7.2. Nonconstant slopes and other readouts	20
References	21

1. INTRODUCTION

A finite representation of a present observation need not be a finite representation of the predictions that observation will support. Expansion separates initially close states, and an interval

Date: Fifth revision, September 22, 2026.

2020 Mathematics Subject Classification. Primary 37D35, 94A34; Secondary 37C45, 60J05, 62B05, 93E11.

Key words and phrases. finite-state prediction, Gibbs measure, expanding repeller, pressure, functional quantization, partial refreshment, noisy acquisition.

without new data forces an observer to retain information about several future readouts. There are consequently two questions. One asks for the geometry of the weighted future orbit. The other asks whether a code for that geometry admits a closed update on a finite persistent state space. This paper gives matching answers on separated Gibbs repellers and extends the resulting prediction laws to acquisition kernels that retain hidden memory.

1.1. The principal results. Let T be the expanding map defined by finitely many separated $C^{1+\gamma}$ inverse branches, let μ be a Gibbs probability, and suppose an acquisition is followed by k unobserved iterates with probability proportional to q^k . Write φ for the normalized Gibbs potential and ψ for the logarithmic derivative of an inverse branch. The principal results identify the optimal M -state squared prediction risk R_M with a finite cylinder profile $G_M(q)$:

$$R_M \asymp e_M^2(Z_q) \asymp G_M(q), \quad Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}.$$

Here G_M is obtained by successively splitting a cylinder of largest mass-weighted orbit diameter. It is a finite construction from cylinder probabilities and branch derivatives, not a restatement of an optimization over orbit codebooks.

Theorem 2.1 supplies this equivalence with the same state budget, up to constants. Its essential realization lemma is that the entire greedy tree is suffix closed: a full b -ary tree with N leaves needs exactly $(bN - 1)/(b - 1)$ designated states. The proof uses stationarity of the Gibbs law to compare the energy of a word with the energy of any suffix. This is different from simply attaching an age register to each codeword, which in general costs the number of codewords times the truncation depth. A disjoint-tube estimate on the orbit image supplies the matching lower bound for arbitrary Hilbert-space centres and hence for all randomized, time-dependent causal observers.

Theorem 2.2 computes the exponent of this profile. If P denotes full-shift pressure, let

$$P(s_g \varphi + 2s_g \psi) = 0, \quad P(s_t \varphi) + s_t \log q = 0.$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \frac{1 - \max\{s_g, s_t\}}{\max\{s_g, s_t\}}.$$

The two roots distinguish spatial resolution from survival through unobserved expansion. Their pressure balance determines the transition, including nonconstant derivative cocycles and nonuniform Markov branch laws. A variational formula expresses them through entropy, information and expansion rates. Positive-matrix formulas and finite pressure enclosures make the invariant explicit in specialized and general Hölder data, respectively.

Theorem 5.1 addresses dependence across acquisitions. The state transition at an acquisition may retain an arbitrary component of the preceding hidden state, subject to a common invariant law and a uniform minorization. A lower bound is obtained by conditional domination, not by declaring the acquisition to be a regeneration. A resetting observer has exactly the same expected loss as in the independent reference experiment. These two facts transfer the intrinsic pressure law and the noisy expanding-torus law to genuinely dependent experiments. In the latter case old observations can affect the current posterior; no identification of that posterior with a latest-observation posterior is needed.

1.2. Resource convention. The constrained resource is the cardinality of one persistent register. The program, fixed real constants, absolute clock and transient workspace are separate resources. Every data-dependent quantity used after a checkpoint must be encoded in the register. In particular, an acquisition indicator is available to the update kernel but not separately to the post-update decoder. Changing this interface can change exact zero-error cardinalities, as Example 6.5 demonstrates. All principal lower bounds allow stochastic kernels and arbitrary time dependence.

Our upper constructions are stationary finite-state machines. We prove their order optimality, not the exact optimality of stationary policies for every member of the unrestricted class.

1.3. Relation to the literature. Classical quantization, causal coding and dynamical information theory provide different pieces of the problem. They should not be identified by a shared use of the word “memory.”

Linder and Yüksel [6] formulate zero-delay coding of vector Markov sources by controlled quantizers; their encoder and decoder can use transmitted histories or posterior states. Wood, Linder and Yüksel [7, Theorem 3] prove deterministic stationary policy optimality for finite irreducible aperiodic Markov sources. A stationary policy on a belief space is not, by itself, a machine with a fixed finite number of persistent values. Here acquisitions are intermittent and all retained observation-dependent information is charged to that finite number. This explains why a separate suffix-cardinality argument is necessary.

Greedy prefix-tree construction has classical antecedents in Tunstall’s variable-to-fixed coding [10]. The present priority is the mass-weighted *orbit distortion*, not merely word probability. Neither greedy tree construction nor the elementary vertex count is claimed as a new coding principle; the issue is to prove that this particular distortion tree is shift closed and matches the unrestricted causal risk.

Hilbert-space functional quantization is also classical. For example, Luschgy and Pagès [8] derive sharp Gaussian functional-quantization asymptotics from covariance eigenvalue decay. We use finite-centre Hilbert quantization as a converse, but must additionally compute a non-Gaussian dynamical orbit geometry and realize its codes causally. Pressure formulas for spatial quantization on cookie-cutter sets already occur in Roychowdhury [9]. The new calculation concerns the *weighted future orbit*: its partition exponent is the maximum of a spatial pressure and a survival pressure, and its adaptive codes have linear-sized suffix closure. The spatial pressure alone is not presented as a new principle.

Finite-rate control [1, 2, 3] constrains communication over time, whereas dynamical rate distortion [4] constrains information in orbit representations. Those resource models are different from the persistent-cardinality model here. The following comparison records the particular distinctions used in the proofs.

Reference or result	Constrained object	Structural conclusion used for comparison
Linder–Yüksel	Per-time transmitted symbol; vector Markov source	Existence and structure of causal quantizer policies on posterior state spaces.
Wood–Linder–Yüksel, Theorem 3	Zero-delay symbols; finite Markov source	Deterministic stationary policy optimality and finite-horizon comparison.
Luschgy–Pagès	Static finite-centre Hilbert code	Gaussian quantization asymptotics from covariance eigenvalues; no finite-state update constraint.
Roychowdhury	Spatial code for a cookie-cutter measure	Quantization dimension related to thermodynamic temperature; no future survival cost.
Theorems 2.1–2.2	All persistent observer states	Matched cylinder profile, arbitrary-coder converse, linear suffix realization and two-pressure exponent.
Theorem 5.1	Same finite register, dependent acquisitions	Conditional minorization lower bound and exact expected-loss transfer for resetting observers.

1.4. Organization and dependencies. Sections 2–4 state and prove the intrinsic profile and pressure theorem. Section 5 proves the dependent-acquisition comparison. The posterior-orbit

and noisy expanding results needed in these arguments are proved in full in Sections 6–7. Those sections also retain the exact uniform critical crossover for the integer expanding experiment.

This principal-theorem edition ends after that causal-memory chain. The complete edition, compiled from the same core files, also contains the calibrated collision-moment theorem, the exact Gaussian nuisance quotient, all earlier quantitative results and the full foundational appendices. No statement or proof is removed from the complete manuscript by this separate reading edition. The foundational connection is precise: acquired-law quantization provides the converse, the suffix theorem provides the resource-aware causal realization, and thermodynamic pressure computes the resulting invariant. The assumptions of a separated finite-branch repeller and of an invariant minorized acquisition kernel are proved or stated where they are used; they are not replacements for the model-specific hypotheses of a Sinai, hard-particle, or unbounded- operator problem.

2. AN INTRINSIC LAW ON GIBBS REPELLERS

Let $b \geq 2$, let $0 < \gamma \leq 1$, and let $h_1, \dots, h_b$ be increasing $C^{1+\gamma}$ maps of $[0, 1]$ into itself. We assume

$$0 < r_- \leq h'_i(x) \leq r_+ < 1, \quad \text{dist}(h_i([0, 1]), h_j([0, 1])) \geq g > 0 \quad (i \neq j). \quad (2.1)$$

Write $\Sigma = \{1, \dots, b\}^{\mathbb{N}_0}$, let S denote the left shift, and put

$$\pi(\omega) = \lim_{n \rightarrow \infty} h_{\omega_0} \cdots h_{\omega_{n-1}}(0), \quad K = \pi(\Sigma), \quad T = \pi S \pi^{-1}.$$

Strong separation makes π a homeomorphism onto K . In particular, T is a well-defined continuous expanding map on the repeller. Our readout is the ambient coordinate $f(x) = x$.

For a word $w = w_0 \cdots w_{n-1}$, set

$$h_w = h_{w_0} \circ \cdots \circ h_{w_{n-1}}, \quad r_w = \sup_{[0,1]} h'_w, \quad K_w = h_w(K), \quad r_\emptyset = 1.$$

The notation w_k means the suffix $w_k \cdots w_{n-1}$. Choose a Hölder potential φ on Σ , normalized by $P(\varphi) = 0$, and let μ be its shift-invariant Gibbs probability. We use μ also for its image on K , and write $\mu[w] = \mu(K_w)$. Here P is topological pressure on the full shift. Thus, for a constant $C_G \geq 1$,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[w] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [w]), \quad S_n u = \sum_{j=0}^{n-1} u \circ S^j. \quad (2.2)$$

The derivative potential is

$$\psi(\omega) = \log h'_{\omega_0}(\pi(S\omega)). \quad (2.3)$$

It is Hölder and is bounded between $\log r_-$ and $\log r_+$.

Fix $0 < q < 1$. At time zero and at each subsequent acquisition a fresh μ -distributed state is observed exactly. At every other time the hidden state is advanced by T . The acquisition indicators after time zero are independent Bernoulli variables with probability $1 - q$ and are independent of all fresh states. After the update, the decoder receives only the persistent state, its absolute time, and independent coding randomness. In particular, it does not receive the current acquisition indicator as a separate uncharged input. An M -state observer has one persistent register with at most M values. Its Borel update and decoding kernels may be randomized and time dependent; no other data-dependent quantity survives a checkpoint. Define

$$R_M = \inf_{\mathcal{A}} \limsup_{H \rightarrow \infty} \frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E} |X_t - \hat{X}_t^{\mathcal{A}}|^2. \quad (2.4)$$

The infimum is over this entire observer class. Its deterministic stationary subclass will suffice for the upper bounds. Program constants and transient computation are not charged to M .

The weighted orbit and its squared quantization error are

$$Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_j \|Z_q(x) - c_j\|^2 d\mu(x). \quad (2.5)$$

The centres in this definition need not lie on the orbit image.

2.1. A finite cylinder algorithm. For every finite word define

$$V_q(w) = (1-q) \sum_{k=0}^{|w|-1} q^k r_{w_k}^2 + q^{|w|}, \quad a_q(w) = \mu[w] V_q(w). \quad (2.6)$$

Both quantities at the empty word are one. They require only finite cylinder masses and derivatives of finite compositions. Set

$$\rho_q = q + (1-q)r_+^2 < 1. \quad (2.7)$$

View all finite words as the vertices of a full b -ary tree. Starting from the empty word, repeatedly replace a leaf of maximal a_q by its b children. Resolve ties in lexicographic order. For $N \geq 1$, let $\mathcal{P}_N$ be the last leaf partition in this sequence having at most N leaves, and put

$$G_N(q) = \sum_{w \in \mathcal{P}_N} a_q(w). \quad (2.8)$$

This convention also defines $G_N = 1$ for $N < b$. No optimization over Hilbert-space centres or over observer policies enters the algorithm.

Theorem 2.1 (Intrinsic profile and finite-state realization). *Under (2.1)–(2.2), for every $0 < q < 1$ there are constants $0 < c \leq C < \infty$ such that, for every integer $M \geq 1$,*

$$cG_M(q) \leq e_M^2(Z_q) \leq R_M \leq CG_M(q). \quad (2.9)$$

The lower bound holds also with the liminf time-average loss of any individual randomized, time-dependent observer. The upper bound is attained to the displayed factor by a deterministic stationary observer which, at each acquisition, selects a leaf of a greedy partition and thereafter deletes the first symbol of its stored word. For every N , the full greedy tree is suffix closed and the constructed observer has exactly $(b|\mathcal{P}_N| - 1)/(b - 1)$ designated states and average loss at most $G_N(q)$. In particular, it uses at most $(bN - 1)/(b - 1) \leq 2N - 1$ states.

The same order $G_M(q)$ holds when the objective in (2.4) is replaced by $\inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E}|X_t - \hat{X}_t^{\mathcal{A}}|^2$. Constants may depend on the branch, distortion, Gibbs and separation data and on q . They can be chosen uniformly on compact q -intervals in $(0, 1)$ when those data are uniformly controlled.

There are two separate assertions here. The cylinder algorithm computes the relevant distortion profile; its suffix closure computes an implementable observer. The number of suffix states is proportional to the number of leaves, not to the number of leaves times their maximum depth. This is the point at which a geometric code becomes a causal code without a growing multiplicative overhead.

2.2. The pressure law. Let s_q and $s_t(q)$ be the unique numbers in $(0, 1)$ satisfying

$$P(s_g \varphi + 2s_g \psi) = 0, \quad P(s_t(q) \varphi) + s_t(q) \log q = 0, \quad s_*(q) = \max\{s_g, s_t(q)\}. \quad (2.10)$$

Existence and uniqueness are part of the next theorem.

Theorem 2.2 (Pressure determines the causal exponent). *In the setting of Theorem 2.1,*

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_q)}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log G_M(q)}{\log M} = \frac{1 - s_*(q)}{s_*(q)}. \quad (2.11)$$

More precisely, for $0 < s < 1$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{|w|=n} a_q(w)^s = \max\{P(s\varphi + 2s\psi), P(s\varphi) + s \log q\}. \quad (2.12)$$

Consequently the transition occurs at

$$q_c = \exp\{-P(s_g\varphi)/s_g\} \in (0, 1). \quad (2.13)$$

For $q \leq q_c$ the exponent is $(1 - s_g)/s_g$; for $q > q_c$ it is $(1 - s_t(q))/s_t(q)$. At the transition, the finite-budget profile is still given, up to constants, by (2.8) and (2.9).

Formula (2.12) is a maximum of two pressures, not pressure evaluated at the pointwise maximum of two potentials. The derivative cocycle enters the first term; survival across unobserved iterates enters the second. The theorem computes the quantization exponent from these data rather than postulating its existence. It does not identify a universal logarithmic correction or an optimal leading constant at q_c for all Gibbs repellers. The finite cylinder profile retains such corrections without discarding them in the exponent limit.

3. CYLINDER GEOMETRY AND CAUSAL REALIZATION

Throughout this section q and the data in (2.1)–(2.2) are fixed. We suppress the subscript q in V and a . A comparison constant never depends on a word or a state budget.

Lemma 3.1 (Distortion and orbit separation). *There is $D \geq 1$ such that*

$$\sup h'_w \leq D \inf h'_w, \quad D^{-1} r_u r_v \leq r_{uv} \leq r_u r_v. \quad (3.1)$$

If $x, y \in K$ have longest common symbolic prefix w , then

$$c_1 V(w) \leq \|Z_q(x) - Z_q(y)\|^2 \leq V(w), \quad c_1 = (1 - q)g^2 D^{-2}. \quad (3.2)$$

For arbitrary words u, v ,

$$V(uv) \leq r_v^2 (V(u) - q^{|u|}) + q^{|u|} V(v), \quad (3.3)$$

$$V(uv) \leq V(u)V(v), \quad a(uv) \leq a(v). \quad (3.4)$$

There exists $0 < a_0 < \rho_q$ such that each one-symbol child wi , $1 \leq i \leq b$, satisfies

$$a_0 a(w) \leq a(wi) \leq \rho_q a(w), \quad \sum_{i=1}^b a(wi) \leq \rho_q a(w). \quad (3.5)$$

Proof. Let H bound the γ -Hölder seminorms of $\log h'_i$. The chain rule, read from the innermost factor, gives

$$|\log h'_w(x) - \log h'_w(y)| \leq H \sum_{j \geq 0} r_+^{j\gamma} \leq H/(1 - r_+^\gamma).$$

This proves the first assertion with $D = \exp\{H/(1 - r_+^\gamma)\}$. The upper multiplicative bound follows from the chain rule; for the lower bound evaluate the inner derivative at a maximizer and bound the outer derivative below by r_u/D .

Put $n = |w|$. For $k < n$, the points $T^k x, T^k y$ belong to the same suffix cylinder. Their distance is at most r_{w_k} . Immediately after that suffix they belong to two different first-level cylinders, so the mean value theorem also bounds this distance below by gr_{w_k}/D . At $k = n$ their distance is at least g . The remaining squared distances are at most one. Multiplying by $(1 - q)q^k$ and summing proves (3.2), including the term q^n by means of its first summand $(1 - q)q^n g^2$. The same upper bound $V(w)$ holds for any two points of K_w .

Split the defining sum for $V(uv)$ at $|u|$ and apply (3.1) to its first part to obtain (3.3). Each suffix derivative r_{v_k} is at least r_v , since the omitted outer maps are contractions. Consequently

$V(v) \geq r_v^2$. Equation (3.3) therefore gives $V(uv) \leq V(u)V(v)$. Since $V(u) \leq 1$ and shift invariance gives

$$\mu[uv] \leq \mu(S^{-|u|}[v]) = \mu[v],$$

we obtain the exact suffix domination $a(uv) \leq a(v)$. No independence of the symbols is used.

For a one-symbol child, $V(i) = q + (1 - q)r_i^2 \leq \rho_q$; submultiplicativity gives $V(wi) \leq \rho_q V(w)$. Conversely, split at $|w|$ and apply the lower distortion estimate to the old summands. Since $V(i) \geq r_i^2 \geq r_-^2$,

$$V(wi) \geq D^{-2}r_-^2(V(w) - q^{|w|}) + r_-^2q^{|w|} \geq D^{-2}r_-^2V(w).$$

The Gibbs bounds imply a positive lower bound β on $\mu[wi]/\mu[w]$, uniformly in w, i : divide the lower bound for $[wi]$ by the upper bound for $[w]$ at the same point, and use $\varphi \geq \inf \varphi$. Take $a_0 = \beta D^{-2}r_-^2/2$, decreased if necessary so that $a_0 < \rho_q$. Finally the children partition $[w]$; summing $\mu[wi]V(wi) \leq \rho_q \mu[wi]V(w)$ proves the final assertion. $\square$

Lemma 3.2 (Balanced greedy partitions). *Let $L_N = |\mathcal{P}_N|$. Then $N/b \leq L_N \leq N$. For $N \geq b$, let θ_N be the energy of the last leaf split in constructing $\mathcal{P}_N$. Every $w \in \mathcal{P}_N$ satisfies*

$$a_0\theta_N \leq a(w) \leq \theta_N, \quad a_0L_N\theta_N \leq G_N \leq L_N\theta_N. \quad (3.6)$$

For every fixed $A \geq 1$ there is $c_A > 0$ such that

$$c_AG_N \leq G_{\lceil AN \rceil} \leq G_N \quad (N \geq 1). \quad (3.7)$$

Moreover, for each $c > 0$, the number of all finite words v with $a(v) \geq c\theta_N$ is at most C_cN .

Proof. The leaf counts are $1 + j(b - 1)$. A maximal energy selected for splitting never increases, since each new child has at most ρ_q times its parent's energy. A leaf still present at the last split was born from a parent of energy at least θ_N , and every current leaf has energy at most θ_N . This proves (3.6); the root-only cases are kept separate. The sum G_N is nonincreasing by (3.5).

For completeness, the reverse comparison in (3.7) does not require a preassigned power law. Choose an integer j with $b^{j+1} > b(A + 1)$. If $\theta_{\lceil AN \rceil} < a_0^{j+1}\theta_N$, every descendant through depth j of each old leaf would have been split: all these nodes have energy at least $a_0^{j+1}\theta_N$. The new partition would then have at least $b^{j+1}L_N$ leaves, which exceeds $\lceil AN \rceil$. Thus $\theta_{\lceil AN \rceil} \geq a_0^{j+1}\theta_N$. Combining this with (3.6) proves the reverse comparison for $N \geq b$. Finitely many smaller budgets can be absorbed into c_A ; their energies are positive.

The finite full tree ending at $\mathcal{P}_N$ has at most $bL_N/(b - 1)$ vertices. Every other vertex descends from a leaf. At depth j below a leaf its energy is at most $\rho_q^j\theta_N$. Only a bounded number of additional *generations* can therefore have energy at least $c\theta_N$; each has at most b^j descendants per leaf. This bounds the total number of such words by C_cN . $\square$

Lemma 3.3 (Quantization of the orbit image). *There are $0 < c_2 \leq C_2 < \infty$ such that*

$$c_2G_M \leq e_M^2(Z_q) \leq C_2G_M \quad (M \geq 1).$$

The lower bound allows arbitrary off-image centres in ℓ^2 .

Proof. Choosing one orbit point from each cylinder of $\mathcal{P}_M$ uses at most M centres and has squared error at most G_M , by the cylinder diameter estimate.

For the reverse bound take $\mathcal{P}_{4bM}$. It has at least $4M$ leaves. For each leaf w let

$$U_w = \{z \in \ell^2 : \text{dist}(z, Z_q(K_w)) < \delta\sqrt{V(w)}\}, \quad 0 < \delta < \sqrt{c_1}/(2\sqrt{2}).$$

These open sets are pairwise disjoint. Indeed, two different leaves have a common prefix u strictly shorter than either leaf. Any points in their two orbit cylinders have distance at least $\sqrt{c_1V(u)}$, whereas $V(w), V(v) \leq 2V(u)$. The sum of the two tube radii is smaller than this separation. Thus M centres meet at most M of the tubes. Every unoccupied cylinder contributes at least $\delta^2a(w)$ to

the distortion. By (3.6), the sum over the unoccupied leaves is at least a fixed multiple of G_{4bM} . Lemma 3.2 compares this with G_M . The argument uses distances to entire cylinders, so it does not require any centre to belong to the orbit image. The finitely many root-only cases can again be absorbed: the measure is nonatomic and the orbit's first coordinate separates distinct states, so each fixed finite quantization error is positive. $\square$

Lemma 3.4 (Exact suffix closure of the greedy tree). *Let $\mathcal{T}_N$ be the set of all vertices, including the root, in the greedy tree with leaf set $\mathcal{P}_N$, and write $L_N = |\mathcal{P}_N|$. Then $\mathcal{T}_N$ is closed under deletion of the first letter and*

$$|\mathcal{T}_N| = \frac{bL_N - 1}{b - 1} \leq \frac{bN - 1}{b - 1}. \quad (3.8)$$

In particular, every suffix of every leaf is already a vertex of this same tree. No additional age states are required.

Proof. Full support and stationarity give, for every nonempty u ,

$$\mu[uv] < \sum_{|z|=|u|} \mu[zv] = \mu[v].$$

Together with $V(uv) \leq V(v)$, this yields the strict inequality $a(uv) < a(v)$. Suppose a vertex w is selected for splitting. Every proper suffix v of w was already an internal vertex at that time. Otherwise some prefix z of v would be a leaf of the current tree, and the decreasing child energies would give $a(z) \geq a(v) > a(w)$, contradicting the maximality of w . Thus the internal vertices are suffix closed.

If w is a leaf, the parent of each nonempty proper suffix of w is a suffix of the parent of w . This parent is internal by the preceding argument (with the empty word interpreted as the root). Its children are vertices of the tree, so the suffix itself is also a vertex. The root and root-only tree cause no exception. This proves closure for all vertices. Finally, a finite full b -ary tree with I internal vertices has bI edges and $I + L_N$ vertices. Hence $bI = I + L_N - 1$, giving (3.8). $\square$

Proof of Theorem 2.1. The posterior-orbit converse, Theorem 6.2, applied to exact observations and geometric cycles, gives $e_M^2(Z_q) \leq R_M$, with a liminf bound for each observer. Its randomization representation is Lemma 6.1. Equivalently, at each acquisition one freezes independent coding randomness and the past; the M possible post-update states supply at most M no-acquisition decoder strings. Quantization bounds their weighted squared loss below, and the disjoint last-acquisition events give the time-average bound. This does not restrict the competing observers to symbolic codes.

For achievability, use $\mathcal{T}_N$ as the state set. At an acquisition x , write its unique leaf word. On a nonacquisition update delete its first letter; the empty word is absorbing until the next acquisition. Decode a nonempty word v by $\pi(v111\cdots)$ and the empty word by $\pi(111\cdots)$. At age $k < |w|$ after acquisition in K_w , the squared error is at most $r_{w_k}^2$; after exhaustion it is at most one. The average squared loss is at most G_N . Every piece of information used by this rule is in its suffix state. Neither reset age nor a precision index is supplied from outside the register.

By Lemma 3.4, this machine uses exactly $(bL_N - 1)/(b - 1)$ designated states and at most C_0N states, with $C_0 = b/(b - 1)$. Take $N = \lfloor M/C_0 \rfloor$ for sufficiently large M . The fixed-dilation comparison gives $G_N \leq CG_M$. For the remaining finitely many budgets use the one-state constant observer and enlarge C . Lemma 3.3 proves the rest of (2.9).

Let d_k be the expected age- k loss of the constructed machine under one acquisition. At time t its expected loss is

$$\sum_{k < t} (1 - q)q^k d_k + q^t d_t \leq \frac{1}{1 - q} \sum_{k \geq 0} (1 - q)q^k d_k.$$

The right side is at most $G_N/(1-q)$. The supremum objective is bounded below by the average objective, proving its stated order. When q stays in a compact subinterval of $(0, 1)$ and the branch and Gibbs data are controlled, ρ_q stays below one and the separation and child-energy constants have common bounds. All constants in the proof can then be chosen uniformly. $\square$

4. PRESSURE, DISTORTION AND SURVIVAL

We use the Gibbs and variational principles on a finite full shift in their usual Hölder form [5, Chapters 1–2]. For a Hölder function u , these principles give a shift-invariant probability ν_u and $C_u \geq 1$ with

$$C_u^{-1} e^{S_n u(\omega) - nP(u)} \leq \nu_u[w] \leq C_u e^{S_n u(\omega) - nP(u)}, \quad P(u) = \sup_{\nu \in \mathcal{M}_S} \left(h_\nu(S) + \int u d\nu \right). \quad (4.1)$$

Here $\mathcal{M}_S$ is the set of shift-invariant probabilities. Only these finite-alphabet statements, not a countable-state spectral or suspension theorem, are used below.

Lemma 4.1 (The mixed partition sum). *Fix $0 < s < 1$, and set*

$$A(s) = P(s\varphi + 2s\psi), \quad B_q(s) = P(s\varphi) + s \log q.$$

There are constants $c_s, C'_s > 0$, independent of n , such that

$$c_s \max\{e^{nA(s)}, e^{nB_q(s)}\} \leq \sum_{|w|=n} a_q(w)^s \leq C'_s \sum_{k=0}^n e^{kB_q(s) + (n-k)A(s)}. \quad (4.2)$$

Proof. For a word w of length n and $\omega \in [w]$, the chain rule and bounded distortion give

$$r_{w_k} \asymp e^{S_{n-k}\psi(S^k\omega)}, \quad \mu[w] \asymp e^{S_n\varphi(\omega)},$$

with constants independent of n, k, w . The term $k = 0$ and the terminal term in (2.6), together with (4.1), give the two lower bounds in (4.2).

For the upper bound use $(\sum x_j)^s \leq \sum x_j^s$ for nonnegative x_j . At a fixed splitting time $k < n$, the resulting sum, up to a constant and the factor $(1-q)^s$, is

$$\sum_{|w|=n} \exp \left\{ sS_k\varphi(\omega) + ks \log q + S_{n-k}(s\varphi + 2s\psi)(S^k\omega) \right\}.$$

The variations of $S_k u$ inside any k -cylinder are uniformly bounded for a Hölder u . Since every prefix can be concatenated with every suffix on the full shift, this sum is comparable to the product of the length- k partition sum for $s\varphi + s \log q$ and the length- $(n-k)$ partition sum for $s\varphi + 2s\psi$. Each is comparable to its pressure exponential by (4.1). The terminal term is the same calculation with $k = n$. Summing over k proves the assertion. $\square$

Proof of Theorem 2.2. Both pressures at $s = 0$ equal $\log b$. At $s = 1$ they are negative: $A(1) \leq 2 \log r_+ < 0$ and $B_q(1) = \log q < 0$. There is a uniform positive lower bound on every one-symbol conditional cylinder mass. Since there are at least two symbols, there is consequently $\eta < 1$ such that $\mu[w] \leq \eta^{|w|}$. Equation (2.2) implies $S_n\varphi \leq n \log \eta + \log C_G$. Integration against any invariant probability gives $\int \varphi d\nu \leq \log \eta < 0$. The variational principle now shows that both A and B_q are strictly decreasing, with their slopes bounded above by a negative constant. Their unique zeros therefore lie in $(0, 1)$. Equation (4.2) proves (2.12): the sum on its right has at most $n+1$ terms, each at most the exponential of $n \max\{A(s), B_q(s)\}$.

We prove the exponent without assuming any quantization dimension. Let $s > s_*(q)$, $s < 1$. Both pressures are negative, so (4.2) is summable in n . Hence

$$H_s := \sum_{w \text{ finite}} a_q(w)^s < \infty.$$

Balance in the greedy tree yields $L_N(a_0\theta_N)^s \leq H_s$, and thus

$$G_N \leq C_s'' N^{-(1-s)/s}. \quad (4.3)$$

For the reverse estimate choose a Gibbs measure ν_* for a zero-pressure potential as follows. When $s_* = s_g$, use $s_g\varphi + 2s_g\psi$; otherwise use $s_t(q)\varphi + s_t(q)\log q$. The spatial summand or the terminal summand of V_q , respectively, shows that

$$\nu_*[w] \leq C_* a_q(w)^{s_*} \quad \text{for every word } w.$$

Since $\mathcal{P}_N$ is a cylinder partition, $1 \leq C_* L_N \theta_N^{s_*}$. Therefore

$$G_N \geq c_* N^{-(1-s_*)/s_*}. \quad (4.4)$$

Let s decrease to s_* in (4.3) and use (4.4). This proves the exponent limit for G_N , and Theorem 2.1 proves the other limits. Finally $P(s_g\varphi) > 0$ because $s_g < 1$ and $P(\varphi) = 0$. Thus the number q_c in (2.13) lies in $(0, 1)$. Strict monotonicity shows that $s_t(q) \leq s_g$ precisely when $B_q(s_g) \leq 0$, which is equivalent to $q \leq q_c$. $\square$

Corollary 4.2 (Entropy–expansion ratios). *Put $I_\nu = -\int \varphi d\nu$ and $\chi_\nu = -\int \psi d\nu$. Then*

$$s_g = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + 2\chi_\nu}, \quad s_t(q) = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + \log(1/q)}. \quad (4.5)$$

All denominators are strictly positive. The quantities χ_ν are the mean expansion rates of T in the coding coordinates.

Proof. The preceding proof gives $I_\nu \geq -\log \eta > 0$, and $\chi_\nu \geq -\log r_+ > 0$. By the variational principle, $A(s) = \sup_\nu \{h_\nu - s(I_\nu + 2\chi_\nu)\}$. Its zero is the supremum of the displayed ratios. For example, at that supremum all the affine expressions are nonpositive, while ratios approaching the supremum make their values approach zero, since the denominators are uniformly bounded above as well as below. The second formula follows in the same way from B_q . The chain rule identifies $-S_n\psi$ with the logarithmic expansion of T^n on its cylinder. $\square$

Proposition 4.3 (Finite pressure enclosures). *For a Hölder potential u , define*

$$Z_n^-(u) = \sum_{|w|=n} \exp\{\inf_{[w]} S_n u\}, \quad Z_n^+(u) = \sum_{|w|=n} \exp\{\sup_{[w]} S_n u\}.$$

If $\text{var}_j(u) = \sup\{|u(x) - u(y)| : x_{0:j} = y_{0:j}\}$, then

$$\frac{1}{n} \log Z_n^-(u) \leq P(u) \leq \frac{1}{n} \log Z_n^+(u), \quad \frac{1}{n} \log \frac{Z_n^+(u)}{Z_n^-(u)} \leq \frac{1}{n} \sum_{j \geq 1} \text{var}_j(u). \quad (4.6)$$

Here $x_{0:j} = y_{0:j}$ means equality of the first j symbols. Thus certified cylinder bounds on the potentials give convergent finite enclosures for the two pressure roots and for q_c .

Proof. Concatenation gives $Z_{n+k}^- \geq Z_n^- Z_k^-$ and $Z_{n+k}^+ \leq Z_n^+ Z_k^+$. Their normalized logarithms have the same limit $P(u)$, since the oscillation of $S_n u$ in a cylinder is at most $\sum_{j=1}^n \text{var}_j(u)$. Superadditivity and subadditivity prove the enclosure, and the same oscillation bound gives its width. Strict monotonicity of the pressures then converts pressure enclosures into root enclosures. $\square$

These are finite enclosures in the intrinsic potentials. As usual with arbitrary real-valued smooth input data, an effective numerical algorithm requires certified access to those data. Neither the resource model nor this proposition asserts a Turing complexity bound for an unspecified real function.

4.1. Markov branch weights and unequal multipliers.

Corollary 4.4 (A Perron–Frobenius formula). *Suppose $h_i(x) = c_i + r_i x$ and the symbols have a stationary Markov law with a strictly positive stochastic matrix $P = (p_{ij})$. Then s_g and $s_t(q)$ are the unique roots*

$$\rho((p_{ij}^{s_g} r_i^{2s_g})_{ij}) = 1, \quad q^{s_t(q)} \rho((p_{ij}^{s_t(q)})_{ij}) = 1. \quad (4.7)$$

In particular,

$$q_c = \rho((p_{ij}^{s_g})_{ij})^{-1/s_g}.$$

For independent symbols with probabilities $p_i > 0$, these equations reduce to $\sum_i (p_i r_i^2)^{s_g} = 1$ and $q^{s_t} \sum_i p_i^{s_t} = 1$.

Proof. Take $\varphi(\omega) = \log p_{\omega_0 \omega_1}$ and $\psi(\omega) = \log r_{\omega_0}$. The Markov cylinder probability has a stationary initial factor and one fewer transition than $e^{S_n \varphi}$, so the Gibbs ratio is bounded above and below. The pressure of an edge potential $u(i, j)$ is $\log \rho((e^{u(i, j)})_{ij})$, by summing products along words and applying Perron–Frobenius to the positive matrix. This yields (4.7). When all rows equal $(p_i)_i$, the matrix has rank one and its nonzero eigenvalue is the displayed sum. $\square$

The powers of p_{ij} in (4.7) are *entrywise powers*. Replacing them by a matrix power of P would erase precisely the information pressure at issue.

Example 4.5 (A nonlinear variable-expansion family). On $[0, 1]$ take

$$h_1(x) = \frac{x}{5} + \varepsilon x(1 - x), \quad h_2(x) = \frac{3}{4} + \frac{x}{6} + \varepsilon x(1 - x), \quad 0 \leq \varepsilon \leq \frac{1}{100}.$$

Their image intervals are $[0, 1/5]$ and $[3/4, 11/12]$. The derivatives lie, respectively, in $[19/100, 21/100]$ and $[1/6 - 1/100, 1/6 + 1/100]$. Thus all hypotheses, including a common positive separation and distortion bound, hold uniformly. Equip the symbols, for example, with the stationary Markov law

$$P = \begin{pmatrix} 4/5 & 1/5 \\ 7/20 & 13/20 \end{pmatrix}.$$

At $\varepsilon = 0$, Corollary 4.4 gives two explicit positive-matrix equations. At $\varepsilon > 0$, Theorem 2.2 and Proposition 4.3 apply to the genuinely nonconstant derivative potential.

This family is not obtained by a smooth change of coordinate from a map with a common multiplier at every fixed point. The two inverse branches have fixed points with derivatives in the two disjoint intervals just displayed. A C^1 conjugacy with nonzero derivative at those points preserves the fixed-point multipliers, which are their reciprocals. Their inequality is therefore an obstruction to such a conjugacy. There remains a topological full-shift coding, as there must for this repeller class; the assertion concerns the metric expansion, not the absence of symbolic coordinates.

5. ACQUISITIONS WITH PERSISTENT HIDDEN STATE

The preceding realization can be used when acquisition does not regenerate the physical state. The appropriate comparison concerns raw prediction risk: the full-history posterior need not equal the posterior based on the latest observation alone.

Let E and Y be standard Borel spaces, let $T : E \rightarrow E$ be Borel, let $f : E \rightarrow \mathbb{R}^r$ be Borel with $\|f\| \leq F$, and let μ be a probability with $T_{\#}\mu = \mu$. Acquisitions occur at $\tau_0 = 0$ and $\tau_{j+1} = \tau_j + L_j$, where the L_j are independent positive integer random variables of common finite mean ℓ . The acquisition clock is independent of the hidden transition and observation randomizations. Between acquisitions $X_t = TX_{t-1}$. At an acquisition $t > 0$ the transition and observation are specified, in this order, by

$$X_t \mid X_{t-1} = x \sim Q(x, du), \quad Y_t \mid X_t = u \sim K(u, dy). \quad (5.1)$$

At time zero $X_0 \sim \mu$ and $Y_0 \sim K(X_0, \cdot)$. We assume

$$\mu Q = \mu, \quad Q(x, A) \geq \epsilon \mu(A) \quad (x \in E, A \text{ Borel}), \quad 0 < \epsilon \leq 1. \quad (5.2)$$

The whole state X_{t-1} can influence X_t through the remaining part of Q . No regeneration indicator is supplied to the observer. The observer sees the acquisition indicators and the Y_t only when acquisitions occur; the persistent-state convention is that of Section 2.

Use a single pair $(U, Y) \sim \mu(du)K(u, dy)$ to define

$$\begin{aligned} s_k &= \mathbb{P}(L_0 > k), \quad w_k = s_k/\ell, \quad W(n) = \sum_{k \geq n} w_k, \\ m_k(y) &= \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0}, \\ B_0 &= \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \end{aligned} \quad (5.3)$$

Let R_M^{dep} denote the infimum of limsup average *raw* squared loss in this dependent experiment.

Theorem 5.1 (Minorized acquisition comparison). *Under (5.1)–(5.2),*

$$R_M^{\text{dep}} \geq \epsilon(B_0 + e_M^2(Z)) \quad (M \geq 1). \quad (5.4)$$

The same lower bound holds for the liminf average loss of each randomized, time-dependent M -state observer. For every $N, n \geq 1$ with $Nn + 1 \leq M$,

$$R_M^{\text{dep}} \leq B_0 + e_N^2(Z^{<n}) + F^2 W(n). \quad (5.5)$$

A one-state zero decoder always has risk at most F^2 . More generally, every observer that overwrites its entire state using only the latest observation at each acquisition has exactly the same sequence of expected one-time losses as in the independent-acquisition experiment with law μK and the same clock, provided it is initialized in the same way.

Proof. Condition on the entire acquisition-indicator sequence. At time zero the hidden marginal is μ . Each intervening transition is either T or Q , both of which preserve μ . Induction therefore shows that the hidden state at each acquisition has marginal μ , even after this conditioning. Its observation pair has law μK . Future survival for k steps has probability s_k , independent of that pair. Thus an observer which restarts from the current observation has the same age-dependent expected loss as in the independent experiment, and hence the same expected one-time loss after averaging over the common clock. This proves the last assertion. Applying the $Nn + 1$ -state construction of Theorem 6.2 proves (5.5); the constant decoder proves the remaining upper bound.

For the converse fix n . At an acquisition condition on the entire hidden and observed past immediately before the transition, and use the independent uniform representation of the observer's kernels from Lemma 6.1. Freeze its coding uniforms, including those used on the hypothetical next n nonacquisition steps. The M possible post-update states determine at most M length- n decoder strings. Conditional on this past, the new observation pair has law $Q(x, du)K(u, dy)$, which dominates $\epsilon \mu(du)K(u, dy)$. Squared loss is nonnegative, so its expected contribution up to the next acquisition, truncated at n , is at least

$$\epsilon \inf_{i(\cdot), c_{i,k}} \sum_{k < n} s_k \mathbb{E}_{\mu K} \|f(T^k U) - c_{i(Y),k}\|^2 = \epsilon \ell (B_{0,<n} + e_M^2(Z^{<n})).$$

For the equality, condition on Y and use orthogonality around $m_k(Y)$; the remaining infimum is precisely finite-dimensional quantization. Enlarging the possible codebooks to all vectors only weakens this lower bound. The encoder is permitted to depend on the conditioned past, but the resulting bound does not.

For a horizon H , sum these truncated contributions only over acquisitions at times at most $H - n$. Each contribution stops at the next acquisition, so distinct terms charge disjoint time

indices, path by path. The expected number of these acquisitions, divided by H , converges to $1/\ell$ by the elementary renewal theorem. Taking $\liminf$ in H proves the lower bound with $B_{0,<n}$ and $Z^{<n}$. Finally let $n \rightarrow \infty$. The noise terms converge by boundedness. The quantization terms increase to $e_M^2(Z)$: projection gives one inequality, and extending a truncated codebook by zero gives

$$e_M^2(Z) \leq e_M^2(Z^{<n}) + \mathbb{E}\|Z - Z^{<n}\|^2 \leq e_M^2(Z^{<n}) + F^2W(n).$$

This proves (5.4). $\square$

The reference number B_0 in this theorem is *not* asserted to be the Bayes risk conditional on the entire dependent history. Older observations can improve that posterior. The theorem uses minorization to obtain a lower bound and invariant marginals to obtain an upper bound; it does not replace a dependent conditional law by an independent one.

Corollary 5.2 (The pressure law survives partial refreshment). *Take the repeller, Gibbs law and geometric acquisition clock of Section 2, but use any kernel Q satisfying (5.2) at acquisitions. Observe the current state exactly at those times. Then, for each fixed $\epsilon > 0$,*

$$R_M^{\text{dep}} \asymp G_M(q), \quad \lim_{M \rightarrow \infty} \frac{-\log R_M^{\text{dep}}}{\log M} = \frac{1 - s_*(q)}{s_*(q)}.$$

The same order holds for supremum one-time loss. An explicit example is

$$Q(x, du) = (1 - \epsilon)\delta_{Tx}(du) + \epsilon\mu(du). \quad (5.6)$$

For $\epsilon < 1$ the physical state is not regenerated at every acquisition. The phase boundary is governed by the acquisition survival probability q , not by the unobserved physical-refresh survival probability $1 - \epsilon(1 - q)$.

Proof. Exact observations give $B_0 = 0$ and the orbit Z_q . Theorem 5.1 supplies the lower bound $\epsilon e_M^2(Z_q)$. The resetting suffix observer in Theorem 2.1 supplies the upper bound $CG_M(q)$ with the same expected losses in the dependent experiment. The pressure formula and the supremum statement follow from the corresponding results there. Kernel (5.6) preserves μ because T does, and has the displayed minorization. Its transition coin is not an observation. $\square$

Corollary 5.3 (Noisy acquisition with retained hidden memory). *On $\mathbb{T}^d$ let $T_b x = bx \pmod{1}$, with integers $b \geq 2, d \geq 1$, let μ be Haar measure, and use any acquisition kernel Q satisfying (5.2). At each acquisition observe $Y = X + \sigma G \pmod{1}$, where the observation noises are independent standard Gaussian vectors. Predict $\Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d$. Assume $W(n) \leq Aw_n$ for every n and some $A < \infty$. With*

$$a_k = e^{-2\pi^2 \sigma^2 b^{2k}}, \quad v_k = w_k a_k^2, \quad B_0 = d \sum_{k \geq 0} w_k (1 - a_k^2),$$

and Ψ_v, n_M as in (7.4) and (7.6), for all $M \geq 1$ and $\sigma \geq 0$,

$$\epsilon \left(B_0 + \frac{\Psi_v(n_M)}{8Ab^6} \right) \leq R_M^{\text{dep}} \leq B_0 + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (5.7)$$

For geometric acquisitions this also bounds the optimal supremum raw loss. In particular $R_M^{\text{dep}} \asymp B_0 + \Psi_w(n_M)$, with constants independent of M and σ when ϵ is bounded below. The critical-window comparison of Corollary 7.4 remains valid for this dependent experiment, and the explicit restarting suffix observer has the same exact crossover coefficient as there.

Proof. The single-acquisition posterior under μK is $m_k(y) = a_k \Phi(b^k y)$, by (7.3). Lemma 7.2 bounds its orbit quantization error below by $\Psi_v(n_M)/(8Ab^6)$; use (5.4). The suffix observer proving Theorem 7.1 overwrites its state at acquisitions, so its exact expected losses, including the formula (7.9), transfer unchanged. This proves the upper bound and the supremum assertion.

Finally $0 \leq \Psi_w(n) - \Psi_v(n) \leq B_0/d$ proves the raw-profile comparison. Its constants are uniform in the same compact q -windows as in Corollary 7.4; the explicit observer has exactly the same loss, rather than merely comparable loss, so its coefficient limit transfers as well. $\square$

6. POSTERIOR ORBITS AND FINITE-STATE REALIZATION

Let E and $\mathcal{Y}$ be standard Borel spaces, let μ be a probability on E , and let K be a probability kernel from E to $\mathcal{Y}$. Let $T : E \rightarrow E$ and $f : E \rightarrow \mathbb{R}^r$ be Borel, with $\|f\| \leq F < \infty$. At each renewal a fresh pair (U, Y) has law $\mu(du)K(u, dy)$. The observer acquires Y , not necessarily U . Between renewals the hidden state follows T . The successive cycle lengths $L_0, L_1, \dots$ are independent copies of a positive integer-valued random variable L , independent of all acquired pairs, and

$$\ell = \mathbb{E}L < \infty, \quad s_k = \mathbb{P}(L > k), \quad w_k = \frac{s_k}{\ell}, \quad W(n) = \sum_{k \geq n} w_k. \quad (6.1)$$

Thus $\sum_{k \geq 0} w_k = 1$. There is an acquisition at time 0. Subsequent acquisition times are $\tau_j = L_0 + \dots + L_{j-1}$. At time $t = \tau_j + k < \tau_{j+1}$, the target is $f(T^k U_j)$. The renewal indicator is observed, but the length of the current cycle is not announced at its beginning.

An M -state observer retains $I_t \in \{1, \dots, M\}$. Its update is an arbitrary Borel stochastic kernel of the preceding state, absolute time, renewal indicator, and, at a renewal only, the new observation. After the update, the decoder has the state, absolute time, and independent public randomness. It does not separately retain the last observation, the renewal indicator, or the renewal age. Private randomness is allowed, but any data-dependent persistent register, including an age or a random-seed position, is part of I_t . Program descriptions, real constants, and transient computation are not charged to M . This is a persistent-cardinality constraint, not a bound on total computational storage.

Choose simultaneous Borel versions

$$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0} \in \mathcal{H} := \ell^2(\mathbb{N}_0; \mathbb{R}^r). \quad (6.2)$$

A countable union of null sets makes these versions simultaneous; values there may be set to zero. Write

$$e_M^2(Z) = \inf_{c_1, \dots, c_M \in \mathcal{H}} \mathbb{E} \min_{i \leq M} \|Z(Y) - c_i\|_{\mathcal{H}}^2, \quad B = \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \quad (6.3)$$

The number B is the unavoidable error of the complete acquisition history. Let $A_t = t - \max\{\tau_j : \tau_j \leq t\}$, and let Y_t^* be the observation at that last renewal. Define

$$E_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2, \quad (6.4)$$

$$R_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|f(X_t) - \hat{f}_t\|^2. \quad (6.5)$$

Here and below an infimum is over the specified machine class; it need not be attained. All statements about e_M^2 use squared distortion.

Lemma 6.1 (Kernel realization). *The preceding machine class can be realized using a countable family of independent uniform random variables, independent of the experiment. After fixing this family and the history before a renewal, the first n outputs along a continuation with no further renewal belong to a set of at most M vectors in $(\mathbb{R}^r)^n$.*

Proof. For a finite-state transition kernel the inverse cumulative probabilities give a Borel function of the input and one uniform variable. For a general standard Borel output, embed its range as a Borel subset of $[0, 1]$, use the measurable inverse distribution function, and map back; a fixed

default value handles the null exceptional set. Applying this construction to each time-indexed kernel uses only countably many uniforms. The product probability space supplies them jointly. Public uniforms are included in the same construction, with their prescribed sharing between update and decoder. Once they are fixed, the update and decoder are deterministic. Given the state immediately after a renewal, the no-renewal input sequence and absolute times determine the subsequent states and outputs. There are only M possible starting states. The argument uses a fixed time-indexed family, not an uncharged data-dependent position in a random tape. $\square$

Theorem 6.2 (Posterior-orbit converse and realization). *For the experiment just defined,*

$$R_M^{\text{av}} = B + E_M^{\text{av}}, \quad e_M^2(Z) \leq E_M^{\text{av}}. \quad (6.6)$$

In fact every randomized, time-dependent M -state machine satisfies the latter lower bound with its $\liminf$ time-average excess loss. Conversely, for all integers $N, n \geq 1$,

$$E_{Nn+1}^{\text{av}} \leq e_N^2(Z^{<n}) + F^2W(n) \leq e_N^2(Z) + F^2W(n), \quad (6.7)$$

where $Z^{<n}$ is the first n coordinates of Z . The upper bound is realized, up to an arbitrary positive error, by a stationary deterministic machine with states $\{1, \dots, N\} \times \{0, \dots, n-1\} \cup \{\partial\}$. In particular, for $M \geq 2$,

$$e_M^2(Z) \leq E_M^{\text{av}} \leq \inf_{1 \leq n \leq M-1} \left\{ e_{\lfloor (M-1)/n \rfloor}^2(Z) + F^2W(n) \right\}. \quad (6.8)$$

Proof. Conditional on the full acquired history, the renewal indicators, and independent coding randomness, the conditional mean of $f(X_t)$ is $m_{A_t}(Y_t^*)$. Indeed, the current acquired pair is independent of previous cycles, and survival of its cycle is independent of that pair. Conditional orthogonality gives, for every machine,

$$\mathbb{E} \|f(X_t) - \hat{f}_t\|^2 = \mathbb{E} \|f(X_t) - m_{A_t}(Y_t^*)\|^2 + \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2.$$

The Cesàro renewal-reward identity makes the average of the first term tend to B . Bounded rewards and $\mathbb{E}L < \infty$ suffice, including arithmetic or deterministic cycle lengths. This proves the equality in (6.6).

Fix n . At a renewal at time s , condition on the past and then use Lemma 6.1. Along the first n no-renewal steps the outputs are chosen from at most M decoder strings. The new Y has its original marginal law under this conditioning. Since survival to age k has probability s_k , the expected excess reward in those steps is at least

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_i \sum_{k < n} s_k \|m_k(Y) - c_{i,k}\|^2 = \ell e_M^2(Z^{<n}).$$

Randomizing the selection of a string cannot reduce its pointwise minimum. Integrating the conditioned variables preserves the bound. Let $u_s = \mathbb{P}(s \text{ is a renewal})$. Rewards arising from distinct last renewals are disjoint. Summing over $0 \leq s \leq N-n$ therefore gives a lower bound $\ell e_M^2(Z^{<n}) \sum_{s=0}^{N-n} u_s$ for the total excess loss up to time $N-1$. The elementary renewal theorem, in its Cesàro form, gives $N^{-1} \sum_{s < N} u_s \rightarrow 1/\ell$. Consequently the lower limit of average loss is at least $e_M^2(Z^{<n})$.

Truncation cannot increase quantization error. Conversely, extending a codebook for the truncation by zero gives

$$e_M^2(Z^{<n}) \leq e_M^2(Z) \leq e_M^2(Z^{<n}) + F^2W(n).$$

Letting $n \rightarrow \infty$ proves the converse, without a compactness assumption on the orbit image or a restriction to on-image centres.

For the upper bound choose an ε -optimal N -centre codebook for $Z^{<n}$. Coordinatewise projection permits its k th coordinate to be written $\sqrt{w_k}c_{i,k}$ with $\|c_{i,k}\| \leq F$; a zero-weight coordinate is immaterial. At a renewal, choose the first nearest centre, a Borel choice, and set the state to

$(i, 0)$. At a nonrenewal change (i, k) to $(i, k + 1)$ until $k = n - 1$, after which the state is ∂ . The latter state is absorbing between renewals. Decode (i, k) as $c_{i,k}$ and ∂ as zero. The average excess reward is at most $e_N^2(Z^{<n}) + \varepsilon + F^2 W(n)$ by the same renewal-reward identity. This is an actual finite-state update; its age information is precisely the charged second coordinate. Let $\varepsilon \downarrow 0$. Unused states are permitted, which proves the last assertion. $\square$

Corollary 6.3 (Invariance of the quantization exponent). *Suppose $W(n) \leq A \exp(-cn^\beta)$ for some $A, c, \beta > 0$, and*

$$\alpha = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z)}{\log M} \quad \text{exists in } (0, \infty). \quad (6.9)$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log E_M^{\text{av}}}{\log M} = \alpha. \quad (6.10)$$

If $e_M^2(Z) \asymp M^{-\alpha}$, more explicitly

$$c_1 M^{-\alpha} \leq E_M^{\text{av}} \leq C_1 M^{-\alpha} (\log M)^{\alpha/\beta} \quad (M \geq 2),$$

with constants depending on the displayed hypotheses. For geometric renewals with parameter $p > 0$, the exponent conclusion also holds for the infimum of the supremum-in-time excess risk.

Proof. Choose any $s > \alpha$ and $n = \lceil ((s/c) \log M)^{1/\beta} \rceil$. Then $\log \lfloor (M-1)/n \rfloor / \log M \rightarrow 1$, whereas $W(n) \leq AM^{-s}$. Apply (6.8); the two logarithmic bounds converge to α . Under the two-sided power bound the same substitution gives the explicit inequalities, enlarging constants for finitely many small M .

For geometric renewals put $q = 1 - p$, so $w_k = pq^k$. A stationary reset-based machine has a nonnegative age-loss sequence d_k . With an initial acquisition at zero its loss at time t is

$$p \sum_{k < t} q^k d_k + q^t d_t \leq p^{-1} \sum_{k \geq 0} w_k d_k.$$

Use the upper machines in Theorem 6.2; the converse follows because a supremum is at least the lower time-average. $\square$

The corollary is an exponent theorem, not a uniform equivalence of errors for arbitrary orbits. It applies to arbitrary Borel dynamics and arbitrary noisy or partial reset observations. Its geometric invariant is the quantization exponent of the *posterior* future orbit; the tail assumption controls the cost of making a finite codebook causal. Computing that invariant in a particular dynamical system remains a separate mathematical task.

Proposition 6.4 (Exact autonomous realization). *Let a ray mean a sequence $c = (c_k)_{k \geq 0}$ in the ball $\{z \in \mathbb{R}^r : \|z\| \leq F\}$. Among stationary deterministic machines, the optimal average excess risk is*

$$\inf_{\substack{|C| \leq M \\ SC \subseteq C}} \mathbb{E} \min_{c \in C} \sum_{k \geq 0} w_k \|m_k(Y) - c_k\|^2, \quad (Sc)_k = c_{k+1}. \quad (6.11)$$

The reset update may depend on the previous state; allowing that dependence does not alter this infimum.

Proof. For a fixed autonomous no-renewal update a and decoder d , a state i determines the ray $(d(a^k i))_{k \geq 0}$. These at most M rays are closed under S . Any reset rule selects one of them, possibly using the previous state; its reward is no smaller than the pointwise minimum over all the rays. Conversely, for a finite shift-closed set, retain the chosen ray as the state, apply S on a nonrenewal, and decode its first coordinate. The first nearest ray is a Borel reset rule independent of the previous state. Renewal reward proves the formula. This is a characterization of the stated autonomous class, not an assertion that all optimal randomized nonstationary machines are autonomous. $\square$

Example 6.5 (Orbit cardinality does not include an age register). Let $E = \{0, 1\}$, $U = 1$ deterministically, $T(0) = T(1) = 0$, and $f(x) = x$. Use geometric renewals with $0 < p < 1$ and a constant reset observation. The posterior orbit is the single ray $(1, 0, 0, \dots)$, so $e_1^2(Z) = 0$ and $B = 0$. Nevertheless a one-state decoder, having only absolute time and independent public coins, does not know whether a renewal occurred at that time. For every $t \geq 1$ the minimum squared error is $p(1 - p)$, since the target is an independent Bernoulli(p) renewal indicator. Two states suffice for zero error: write 1 at each renewal and 0 otherwise. Thus the orbit converse cannot be reversed with the same cardinality without a realizability condition. The finite-chain overhead in (6.7) and the closure in (6.11) address a genuine obstruction.

7. NOISY EXPANDING EXPERIMENTS AND A UNIFORM CRITICAL WINDOW

The realization overhead in Theorem 6.2 is not always necessary. We now identify a class in which symbolic closure yields matching errors, including the acquisition noise and the renewal tail. Let $b \geq 2$ and $d \geq 1$ be integers, let $\mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$, and put

$$T_b x = bx \pmod{1}, \quad \Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d. \quad (7.1)$$

At a renewal U is Haar uniform and the observation is

$$Y = U + \sigma G \pmod{1}, \quad G \sim N(0, I_d), \quad \sigma \geq 0. \quad (7.2)$$

The noise, the fresh states, and the renewal lengths are mutually independent. The target is $\Phi(X_t) \in \mathbb{R}^{2d}$, not a discontinuous choice of coordinates on the torus. Both average raw risk and average excess risk have the meanings of Section 6.

Haar invariance implies that Y and G are independent: conditional on G , the translate of U is Haar. Consequently the posterior law of U given $Y = y$ is that of $y - \sigma G$ modulo one, and

$$m_k(y) = a_k \Phi(b^k y), \quad a_k = \exp(-2\pi^2 \sigma^2 b^{2k}), \quad B = d \sum_{k \geq 0} w_k (1 - a_k^2). \quad (7.3)$$

This calculation uses the Gaussian characteristic function, so it also specifies exactly which noise information is used by the decoder. Write

$$v_k = w_k a_k^2, \quad \Psi_v(n) = \sum_{k < n} v_k b^{-2(n-k)} + \sum_{k \geq n} v_k, \quad n \geq 0. \quad (7.4)$$

For $v = w$, use the notation Ψ_w . We impose the quantitative renewal condition

$$W(n) \leq A w_n \quad (n \geq 0) \quad (7.5)$$

for some finite $A \geq 1$. In terms of the cycle length this is a uniform bound on the mean residual lifetime, conditional on survival. It includes geometric renewals, mixtures with geometric parameters bounded away from zero, and bounded renewal lengths. It does not include arbitrary polynomial renewal tails.

Put $a = b^d$ and

$$S_n = 1 + a + \dots + a^n, \quad n_M = \max\{n \geq 0 : S_n \leq M\}. \quad (7.6)$$

The distinction between a^n and S_n counts, rather than suppresses, the persistent information about remaining precision.

Theorem 7.1 (Sharp posterior-orbit law). *Under (7.5), for every $M \geq 1$ and $\sigma \geq 0$,*

$$\frac{\Psi_v(n_M)}{8Ab^6} \leq E_M^{\text{av}} \leq \frac{d\pi^2}{3} \Psi_v(n_M), \quad (7.7)$$

$$B + \frac{\Psi_v(n_M)}{8Ab^6} \leq R_M^{\text{av}} \leq B + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (7.8)$$

The lower bounds hold for all randomized, nonstationary machines. The upper bounds are attained by the finite suffix machine described below. Its exact average raw risk with S_n states is

$$U_{n,\sigma} = B + d \sum_{k < n} v_k \left[1 - \text{sinc}^2(\pi b^{-(n-k)}) \right] + d \sum_{k \geq n} v_k, \quad (7.9)$$

where $\text{sinc}(z) = \sin(z)/z$ and $\text{sinc}(0) = 1$. For geometric renewals, (7.8) also holds for $R_M^{\text{sup}} := \inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E} \|\Phi(X_t) - \hat{f}_t\|^2$, with an initial acquisition at zero. The supremum raw risk of the displayed suffix machine is exactly $U_{n,\sigma}$.

Lemma 7.2 (Orbit separation and small balls). *Suppose $v_k = w_k a_k^2$ with a_k nonincreasing and (7.5) holds. For Haar Y the quantization error of $(\sqrt{v_k} \Phi(b^k Y))_{k \geq 0}$ satisfies*

$$e_M^2 \geq \frac{\Psi_v(n_M)}{8Ab^6}. \quad (7.10)$$

Proof. First, $\sum_{k \geq n} v_k \leq Av_n$. Also Ψ_v is nonincreasing, and

$$\Psi_v(n+1) \geq b^{-2} \Psi_v(n). \quad (7.11)$$

These follow directly from (7.4) and monotonicity of a_k . For $x, y \in \mathbb{T}^d$, write $\delta = \max_j \text{dist}_{\mathbb{T}}(x_j, y_j) \in [0, 1/2]$. If $\delta > 0$, set $K = \lfloor \log_b(1/(2\delta)) \rfloor$. In a coordinate attaining δ , for $0 \leq k \leq K$ the unreduced difference has modulus $b^k \delta \leq 1/2$. Hence $|\sin(\pi b^k \delta)| \geq 2b^k \delta$, and the squared orbit distance $D^2(x, y)$ obeys

$$\begin{aligned} D^2(x, y) &= 4 \sum_{k \geq 0} v_k \sum_{j=1}^d \sin^2(\pi b^k (x_j - y_j)) \\ &\geq 16\delta^2 \sum_{k=0}^K v_k b^{2k} \geq \frac{4}{b^2} \sum_{k=0}^K v_k b^{-2(K-k)} \geq \frac{4}{Ab^2} \Psi_v(K). \end{aligned} \quad (7.12)$$

The middle inequality uses $\delta > b^{-(K+1)}/2$; the last uses the residual-tail bound and $A \geq 1$.

Let $h = \lfloor 2(2M)^{1/d} \rfloor^{-1}$ and $K_h = \lfloor \log_b(1/(2h)) \rfloor = \lfloor \log_a(2M) \rfloor$. Since $M < S_{n_M+1}$ and $a \geq 2$, one has $K_h \leq n_M + 2$. Whenever $\delta \geq h$, the associated $K \leq K_h$; thus (7.11) and (7.12) give

$$D^2(x, y) \geq \gamma \Psi_v(n_M), \quad \gamma = \frac{4}{Ab^6}.$$

If $\Psi_v(n_M) = 0$, the conclusion is immediate. Otherwise take a Hilbert ball of radius r , with $r^2 = \gamma \Psi_v(n_M)/16$. Two preimages in the same ball have orbit distance at most $2r$, so their torus sup-distance is less than h . Fixing one preimage, all preimages lie in a torus cube of side $2h$, whose Haar measure is $1/(2M)$. This argument applies even when the centre of the Hilbert ball is not on the orbit image. The union of M such balls has preimage measure at most $1/2$. Every M -centre quantizer therefore has mean squared error at least $r^2/2 = \Psi_v(n_M)/(8Ab^6)$. $\square$

Proof of Theorem 7.1. The lower bound follows from Theorem 6.2, (7.3), and Lemma 7.2. The orthogonality identity then adds B to the excess risk.

For the upper bound use the alphabet $\{0, \dots, b-1\}^d$. The states are all words of lengths $0, \dots, n$, exactly S_n states. At a renewal retain the first n base- b digits of each coordinate of Y , grouped into one word. Use the terminating expansion at ambiguous endpoints; those endpoints have Haar measure zero. At a nonrenewal delete the first letter, leaving the empty word fixed. For a nonempty word w of length l , let $c(w)$ be the midpoint of its b -adic cube, and decode it as

$$a_{n-l} \text{sinc}(\pi b^{-l}) \Phi(c(w)).$$

Decode the empty word as zero. The age before exhaustion is $n-l$, so there is no separate age register. The update does not use the renewal law.

Conditional on this suffix at age $k < n$, the vector $b^k Y$ is uniform on its cube. Averaging each trigonometric coordinate over an interval gives its midpoint value times $\text{sinc}(\pi b^{-(n-k)})$. The displayed output is therefore the conditional mean of $\Phi(b^k U)$ under the suffix. Its raw squared error is

$$d \left[1 - a_k^2 \text{sinc}^2(\pi b^{-(n-k)}) \right].$$

After exhaustion the raw error is d . Renewal reward proves (7.9). For $0 \leq t \leq 1/2$,

$$\frac{4}{3}t^2 \leq 1 - \text{sinc}^2(\pi t) \leq \frac{\pi^2}{3}t^2. \quad (7.13)$$

Indeed, if V, V' are independent uniform variables on an interval of length t , the middle expression is $2\mathbb{E}\sin^2(\pi(V - V'))$. Use $2|u| \leq |\sin(\pi u)| \leq \pi|u|$ for $|u| \leq 1/2$ and $\mathbb{E}(V - V')^2 = t^2/6$. This proves the average upper bounds, including $n = 0$.

For geometric renewals put $q = 1 - p$. The raw age-loss sequence just computed is nondecreasing: a_k decreases and, as k increases before exhaustion, the sinc factor decreases as well. The age at time t has the law of $\min(G_0, t)$, where $\mathbb{P}(G_0 = k) = pq^k$. Therefore its expected raw loss increases to $\sum_k pq^k d_k = U_{n,\sigma}$ and never exceeds this number. The lower bound for supremum loss follows from the average converse. $\square$

Corollary 7.3 (Joint statistical and persistent-state law). *Under the hypotheses of Theorem 7.1,*

$$R_M^{\text{av}} \asymp_{b,d,A} B + \Psi_w(n_M). \quad (7.14)$$

The same holds for raw supremum risk under geometric renewals. Moreover a suffix decoder which does not know σ or the renewal law attains this order of raw risk.

Proof. Every coefficient in Ψ_v is between zero and one. Thus

$$0 \leq \Psi_w(n) - \Psi_v(n) \leq \sum_k w_k(1 - a_k^2) = B/d.$$

This and (7.8) prove the equivalence. For the noise-independent decoder use $\text{sinc}(\pi b^{-l})\Phi(c(w))$, omitting a_{n-l} . Compare with the hypothetical target $\Phi(b^k Y)$. At age k ,

$$\mathbb{E}\|\Phi(b^k U) - \Phi(b^k Y)\|^2 = 2d(1 - a_k) \leq 2d(1 - a_k^2).$$

The squared triangle inequality and the noiseless suffix bound give average raw risk at most $4B + (2d\pi^2/3)\Psi_w(n)$. For geometric renewals the same bound holds for supremum loss: both the noise-discrepancy sequence $1 - a_k^2$ and the noiseless suffix raw-loss sequence are nondecreasing, so each time expectation is bounded by its stationary-age average. The machine uses only b, d, n and the observed digits. $\square$

7.1. Geometric renewals and the critical window. For $w_k = pq^k$, $p = 1 - q$, the noiseless profile is

$$\Psi_q(n) = pb^{-2n} \sum_{k=0}^{n-1} (b^2 q)^k + q^n. \quad (7.15)$$

Theorem 7.1 is uniform for q in any fixed compact subset of $(0, 1)$, since $A = 1/p$ is then bounded. In particular it remains valid when q and σ depend on M .

Corollary 7.4 (Phase regimes and crossover). *For each fixed $0 < q < 1$, at $\sigma = 0$ the optimal average and supremum raw risks have orders*

$$\begin{cases} M^{-2/d}, & q < b^{-2}, \\ (1 + \log M)M^{-2/d}, & q = b^{-2}, \\ M^{-\log(1/q)/(d \log b)}, & q > b^{-2}. \end{cases} \quad (7.16)$$

Let $q_n = b^{-2} \exp(x/n)$ and $H(x) = (e^x - 1)/x$, with $H(0) = 1$. Uniformly for x in bounded intervals,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) \longrightarrow (1 - b^{-2})H(x). \quad (7.17)$$

For the explicit S_n -state suffix machine, its exact noiseless raw risk satisfies the stronger coefficient limit

$$\frac{b^{2n}}{n} U_{n,0} \longrightarrow \frac{d\pi^2}{3} (1 - b^{-2})H(x). \quad (7.18)$$

The optimal risks are bounded above and below by constant multiples of the profile in (7.17), uniformly in this window; no equality of their leading constants is asserted.

Proof. Since $S_n \asymp_b b^{dn}$, evaluating the geometric sum in (7.15) proves (7.16). In the critical window,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) = (1 - q_n) \frac{1}{n} \sum_{k=0}^{n-1} e^{xk/n} + \frac{e^x}{n}.$$

The Riemann sum tends uniformly on bounded x -intervals to $\int_0^1 e^{xt} dt = H(x)$. For the exact suffix risk use $1 - \text{sinc}^2(\pi t) = (\pi^2/3)t^2 + O(t^4)$, uniformly for $0 \leq t \leq 1/2$. After multiplication by b^{2n} , the error sum is bounded by a constant times

$$\sum_{k < n} e^{xk/n} b^{-2(n-k)},$$

which is uniformly bounded on bounded x -intervals. The exhaustion term $db^{2n}q_n^n = de^x$ is bounded as well. Dividing by n proves (7.18). The final assertion is exactly the uniform two-sided bound of Theorem 7.1. $\square$

For completeness, the noise floor has the same transition. For $0 < \sigma \leq 1$, set $n_\sigma = \lfloor \log_b(1/\sigma) \rfloor$; for $\sigma > 1$ set $n_\sigma = 0$. From $1 - e^{-4\pi^2 z^2} \asymp \min(1, z^2)$ and $b^{-(n_\sigma+1)} < \sigma \leq b^{-n_\sigma}$ one obtains

$$B \asymp_{b,d} \Psi_q(n_\sigma), \quad R_M \asymp_{b,d,q} \Psi_q(n_\sigma) + \Psi_q(n_M). \quad (7.19)$$

At $\sigma = 0$ the first term is interpreted as zero. The comparison in the second formula is uniform on compact q -intervals. Thus the small-noise orders of B are respectively σ^2 , $\sigma^2(1 + \log(1/\sigma))$, and $\sigma^{\log(1/q)/\log b}$ in the three regimes. This describes filtering uncertainty and expanding transport in the same experiment, including joint limits, rather than adding an unrelated noisy model.

7.2. Nonconstant slopes and other readouts. The preceding law is not confined to a constant-slope coordinate presentation. The next statement records both a genuine robustness class and its precise boundary.

Proposition 7.5 (Transport under conjugacy and metric readouts). *Let $h : \mathbb{T}^d \rightarrow E'$ be a Borel isomorphism. Transport the reset law, observation (7.2), dynamics, and readout by h ; thus the new observation is $h(Y)$, the dynamics are $hT_b h^{-1}$, and the readout is Φh^{-1} . All the preceding finite-state risks are unchanged. Suppose instead that E' is compact and the new continuous readout g satisfies, for some $0 < c \leq C < \infty$,*

$$c \|\Phi(x) - \Phi(y)\| \leq \|g(hx) - g(hy)\| \leq C \|\Phi(x) - \Phi(y)\|. \quad (7.20)$$

Then its optimal raw squared risks, for the same observations and state budgets, lie between $c^2/4$ and $4C^2$ times those for Φ . This assertion concerns raw risk, not equality of posterior excess risk for nonlinear readouts.

Proof. Apply h^{-1} to each acquired observation within the transient update computation. This and its inverse transform machines without changing a persistent state. For the readout assertion, replace any output z for Φ by a nearest point $\Phi(x_z)$ on its compact image, choosing measurably. For a true state x , $\|\Phi(x_z) - \Phi(x)\| \leq 2\|z - \Phi(x)\|$. Output $g(hx_z)$ and use the upper inequality in (7.20). The inverse construction projects onto the compact image of g and uses the lower inequality. Both transformations act only on the decoder, use no extra states, and are valid conditionally on any coding randomness. $\square$

For example, in one dimension take

$$h_\varepsilon(x) = x + \frac{\varepsilon}{2\pi} \sin(2\pi x) \pmod{1}, \quad |\varepsilon| < \frac{b-1}{b+1}.$$

This is a smooth circle diffeomorphism and the conjugate map has nonconstant derivative

$$(h_\varepsilon T_b h_\varepsilon^{-1})'(h_\varepsilon x) = b \frac{1 + \varepsilon \cos(2\pi bx)}{1 + \varepsilon \cos(2\pi x)} > 1.$$

Coordinatewise products give higher-dimensional examples, with nonuniform physical reset density and transported nonadditive observation noise. The ordinary circle embedding of $h_\varepsilon x$ satisfies (7.20). These are nonconstant-slope expanding experiments, but their Lyapunov exponent remains $\log b$ by conjugacy. We do not identify their law with a pressure theorem for arbitrary nonconjugate expanding maps. The general posterior-orbit theorem still applies to such maps; its quantization invariant would have to be evaluated separately.

REFERENCES

- [1] G. N. Nair and R. J. Evans, Stabilization with data-rate-limited feedback: tightest attainable bounds, *Systems Control Lett.* **41** (2000), 49–56. doi:10.1016/S0167-6911(00)00037-2.
- [2] S. Tatikonda and S. Mitter, Control under communication constraints, *IEEE Trans. Automat. Control* **49** (2004), 1056–1068. doi:10.1109/TAC.2004.831187.
- [3] A. S. Matveev and A. V. Savkin, *Estimation and Control over Communication Networks*, Control Engineering, Birkhäuser, Boston, 2009. doi:10.1007/978-0-8176-4607-3.
- [4] E. Lindenstrauss and M. Tsukamoto, From rate distortion theory to metric mean dimension: variational principle, *IEEE Trans. Inform. Theory* **64** (2018), 3590–3609. doi:10.1109/TIT.2018.2806219. arXiv:1702.05722.
- [5] R. Bowen, *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms*, second revised edition, edited by J.-R. Chazottes, Lecture Notes in Mathematics 470, Springer, Berlin, 2008. doi:10.1007/978-3-540-77695-6.
- [6] T. Linder and S. Yüksel, On optimal zero-delay coding of vector Markov sources, *IEEE Trans. Inform. Theory* **60** (2014), 5975–5991. doi:10.1109/TIT.2014.2346780.
- [7] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* **63** (2017), 5968–5980. doi:10.1109/TIT.2017.2692215.
- [8] H. Luschgy and G. Pagès, Sharp asymptotics of the functional quantization problem for Gaussian processes, *Ann. Probab.* **32** (2004), 1574–1599. doi:10.1214/009117904000000324.
- [9] M. K. Roychowdhury, Quantization dimension for Gibbs-like measures on cookie-cutter sets, *Kyoto J. Math.* **54** (2014), 239–257. doi:10.1215/21562261-2642377.
- [10] B. P. Tunstall, *Synthesis of noiseless compression codes*, Ph.D. dissertation, Georgia Institute of Technology, 1967.